ROUTE INVESTIGATION REPORT FOR NNWADORO PIPELINE PROJECT (ROUTE OPTION 1)

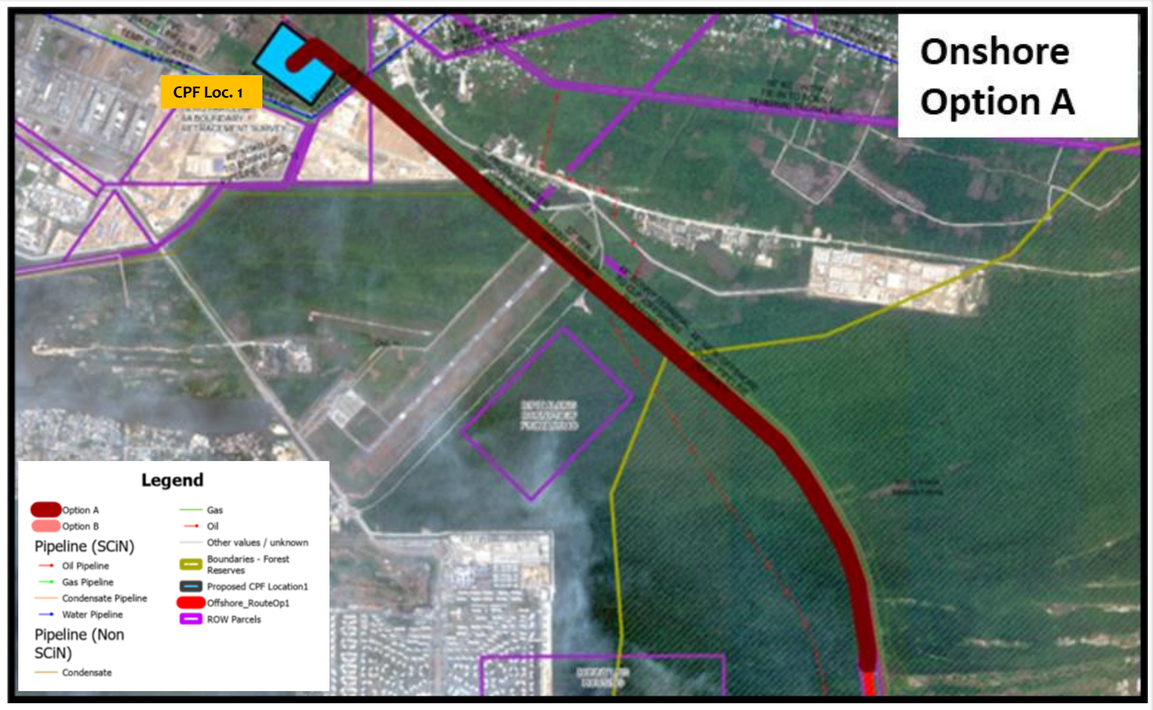
INTRODUCTION:
This project has an opportunity to secure a custom-made financial plan with NNPC to create an investable "Nigeria first" deep water gas development with ~9Tcf of Nnwa-Doro gas and safely supply the NLNG plant with ~700MMscf/d for 20years from 2029. This aligns with the government's decade of gas aspirations and creates immense value to Nigeria and unit parties.

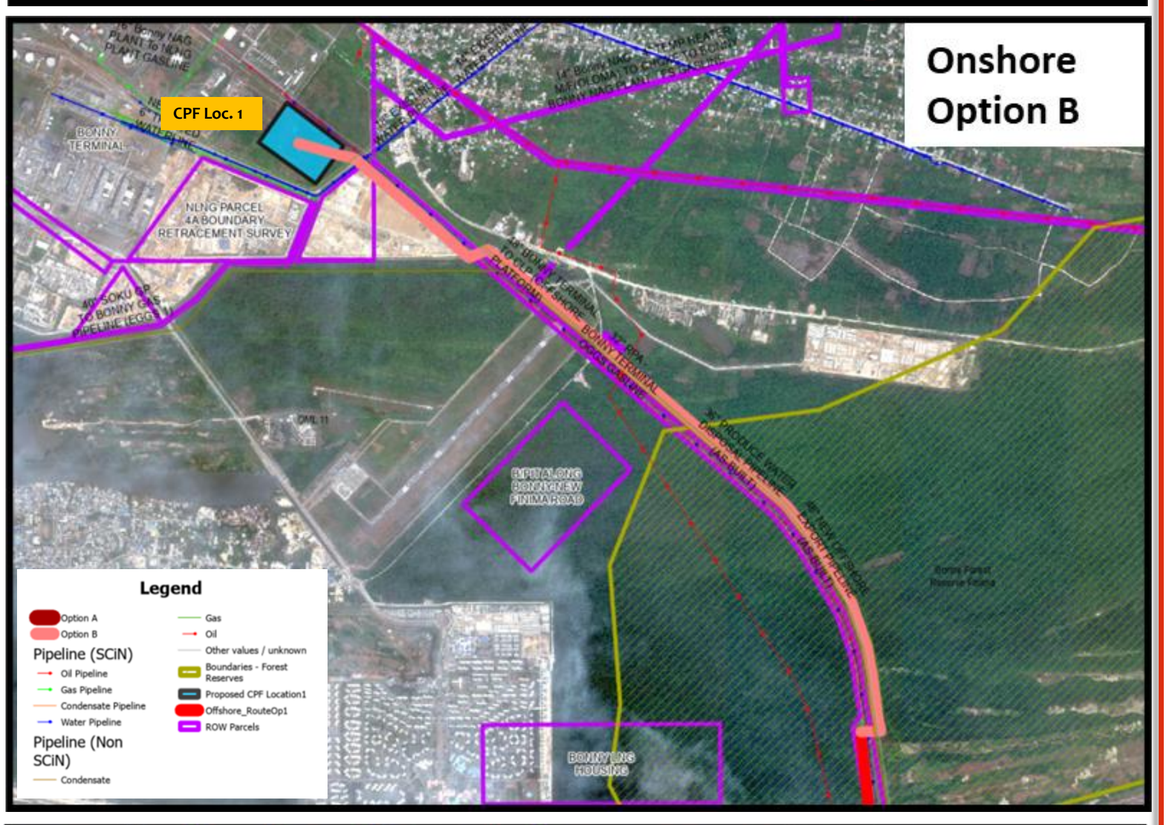
OBJECTIVE:
To identify and propose onshore gas-line route options for Gas evacuation starting from proposed offshore gas-line route option landfall from Nnwadoro field location (OML 135) to NLNG plant in BOGT.

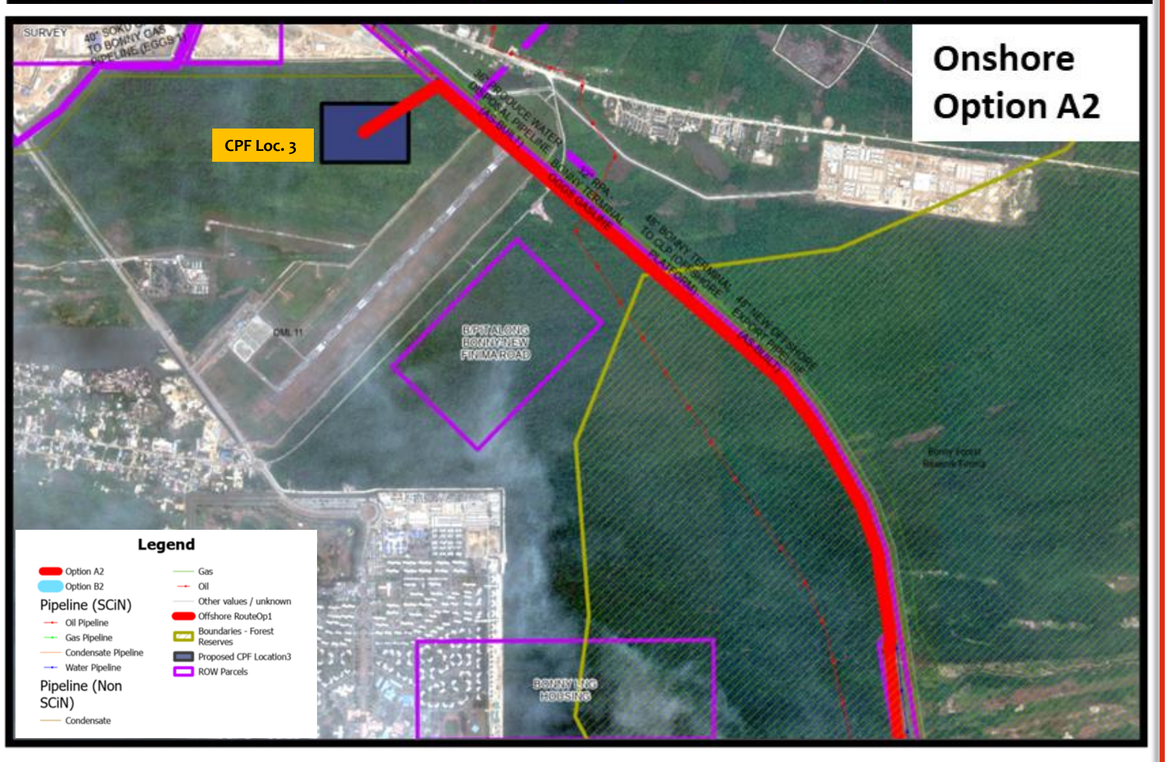
SUMMARY:
From the Route Assessment Statistics for offshore landfall route option 1, a total of four (4) proposed onshore route options were investigated to two (2) proposed CPF locations (2 options per CPF location) in BOGT field location. Proposed Route Option 1A has the shortest route length of 168.25km to CPF location 1 while Proposed Route Option 1A2 has the shortest route length of 166.97km to CPF location 3.

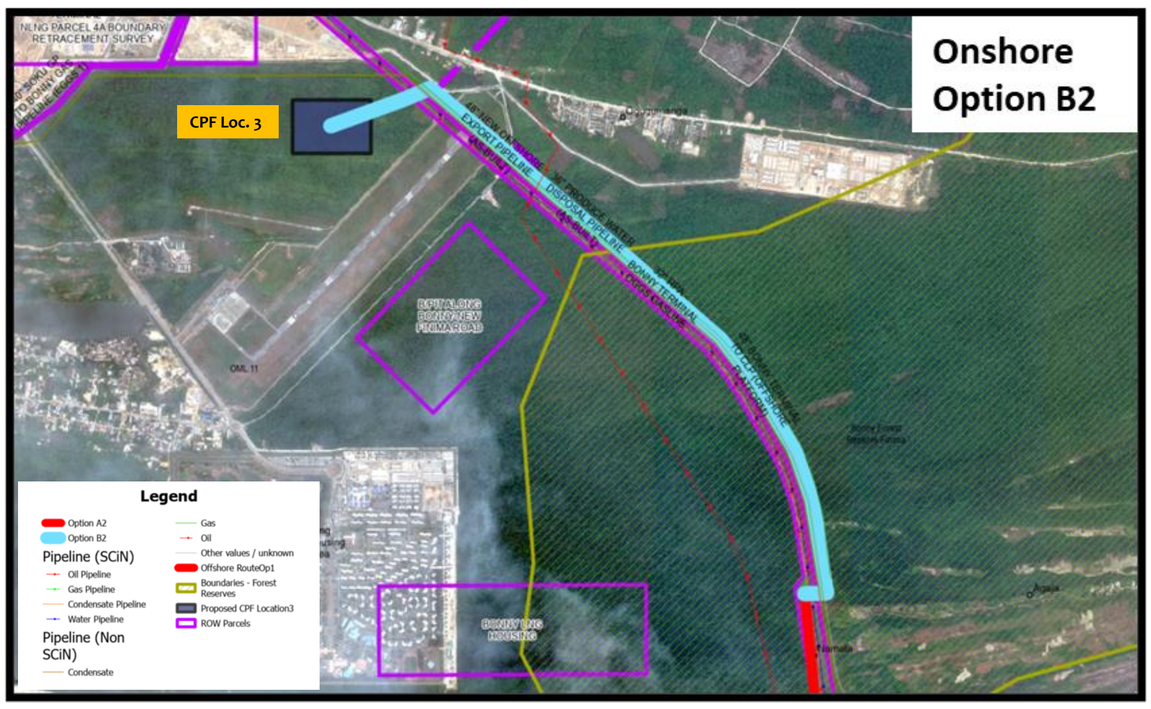
METHODOLOGY:
- Analysis of the most recent high-res satellite imagery for land-use and land cover.
- Considerations for planning the route are environmental, technical, economic and socio-cultural issues using satellite imagery and vector data.
- Generate statistical assessment for the proposed gas line route.

DATASETS USED:
Raster- High Resolution Satellite Imagery of 0.5m resolution acquired January 2023 (Source: SPDC Geomatics SDE).
Vector- The dataset used from Geomatics corporate database are: SPDC Land Parcels Portfolio; Settlement layer data; Electric Powerlines data; Pipelines(SCiN); Flowlines(SCiN); Telecoms All; Gas Plants; Flow Station and Manifold layers.









ROUTE ASSESSMENT STATISTICS FOR NNWADORO PIPELINE

DESCRIPTION	Route Option 1 to CPF Location 1		Route Option 1 to CPF Location 3	
	Option 1A	Option 1B	Option 1A2	Option 1B2
***Approx. total length (km)	168.25	168.43	166.97	167.24
Length Offshore	162.8	162.8	162.8	162.8
***Length Onshore	5.45	5.63	4.17	4.44
***Length through existing SPDC ROW (km)	5.45	0.56	3.70	0.13
***Length through new ROW (km)	Nil	5.07	0.47	4.31
***No. of Forest reserve traversed	1	1	1	1
***No of Communities within 2km buffer	6 (NAMATA, AYAMBO, OGUGUMANGA, ATSAMA, AGAJA, BONNY)	6 (NAMATA, AYAMBO, OGUGUMANGA, ATSAMA, AGAJA, BONNY)	6 (NAMATA, AYAMBO, OGUGUMANGA, ATSAMA, AGAJA, BONNY)	6 (NAMATA, AYAMBO, OGUGUMANGA, ATSAMA, AGAJA, BONNY)
***No. of existing river/creek crossing	Nil	Nil	Nil	Nil
***Width range of river /creek crossings (m)	Nil	Nil	Nil	Nil
***No river /creek crossing with width of less 20m	Nil	Nil	Nil	Nil
***No river /creek crossing with width Greater than 20m	Nil	Nil	Nil	Nil
***No river /creek crossing with width Greater than 50m	Nil	Nil	Nil	Nil
***No river /creek crossing with width Greater than 100m	Nil	Nil	Nil	Nil
No of Flowline in RoW Corridor	Nil	Nil	Nil	Nil
***Pipeline crossings	Several	Several	Several	Several
***No of Telecom(Nitel) cable crossings	Nil	Nil	Nil	Nil
***No of Electrical cable crossings	Nil	Nil	Nil	Nil
***Approx. Length of Farmland traversed (Km)	Nil	Nil	Nil	Nil
***Approx. Length of Impacted Vegetation traversed (Km)	5.45	5.63	4.17	4.44
Approx. Length of Mature Forest traversed (Km)	Nil	Nil	Nil	Nil
Approx. Length of Swamp Vegetation traversed (Km)	Nil	Nil	Nil	Nil
***Approx. Length of Settlement traversed (Km)	3.14	3.18	1.84	2.00
***Major Road Crossing	3	3	1	1

CONCLUSION:
From the route assessment statistics for offshore landfall route option 1, four (4) proposed onshore route options were investigated to two (2) proposed CPF (1 and 3) locations (2 options per CPF location) in BOGT field location for this project. As required, several vital statistics based on the Environmental and Safety, Engineering and Operational, Biodiversity considerations were analyzed and extracted in the Route Assessment Statistical table above. The route options were not ranked in this study but left for ranking during the multi discipline route selection workshop using the SCAN tool. The descriptions on the route assessment statistics table marked (***) are part of the criteria to be used with the SCAN tool for ranking the identified route options during the route selection workshop.

PREPARED BY: Amaechi Ojeh (SPDC-UPV/C/NGI) **CLIENT: Tarryla Nnoaham (SPDC-UPV/C/NF)** **ONSHORE OPs REP: Seember Gbargbar (SPDC-UPV/C/NGO)** **REVIEWED BY: Simisola Oyetunde (SPDC-UPV/C/NGI)**

DATE: 2nd June 2023

ROUTE INVESTIGATION REPORT FOR NNWADORO PIPELINE PROJECT (ROUTE OPTION 2)

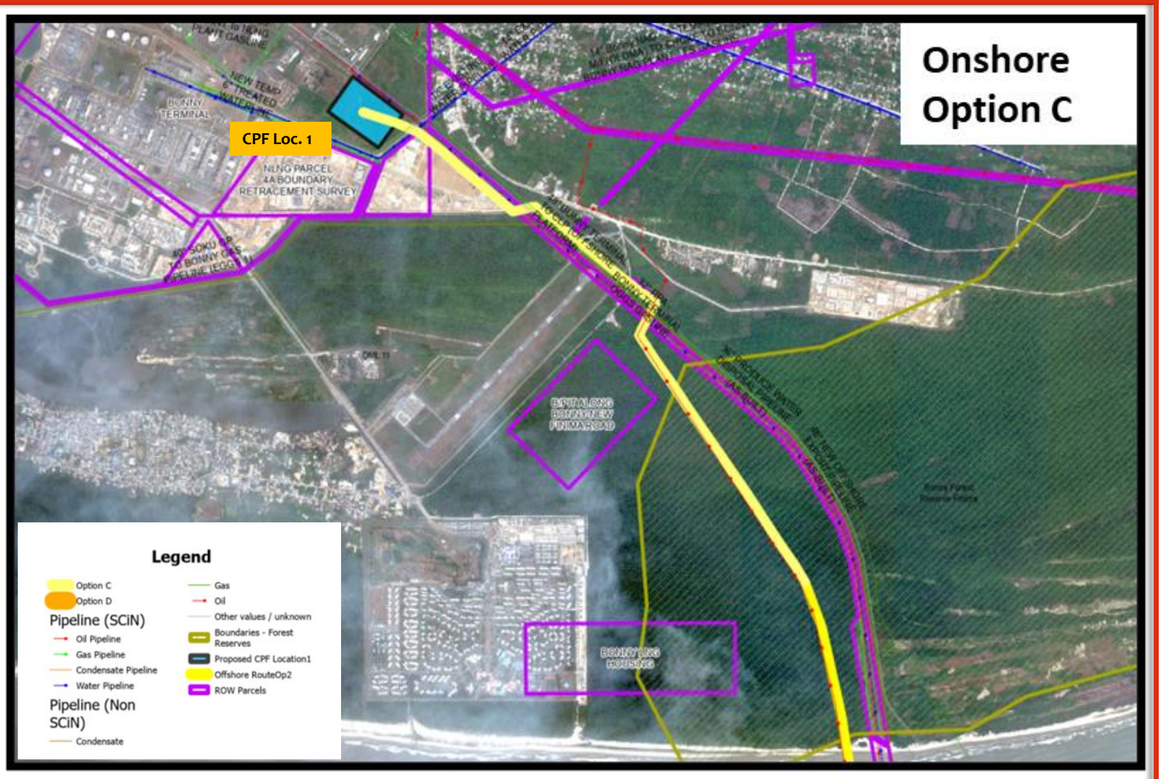
INTRODUCTION:
This project has an opportunity to secure a custom-made financial plan with NNPC to create an investable "Nigeria first" deep water gas development with ~9Tcf of Nnwa-Doro gas and safely supply the NLNG plant with ~700MMscf/d for 20years from 2029. This aligns with the government's decade of gas aspirations and creates immense value to Nigeria and unit parties.

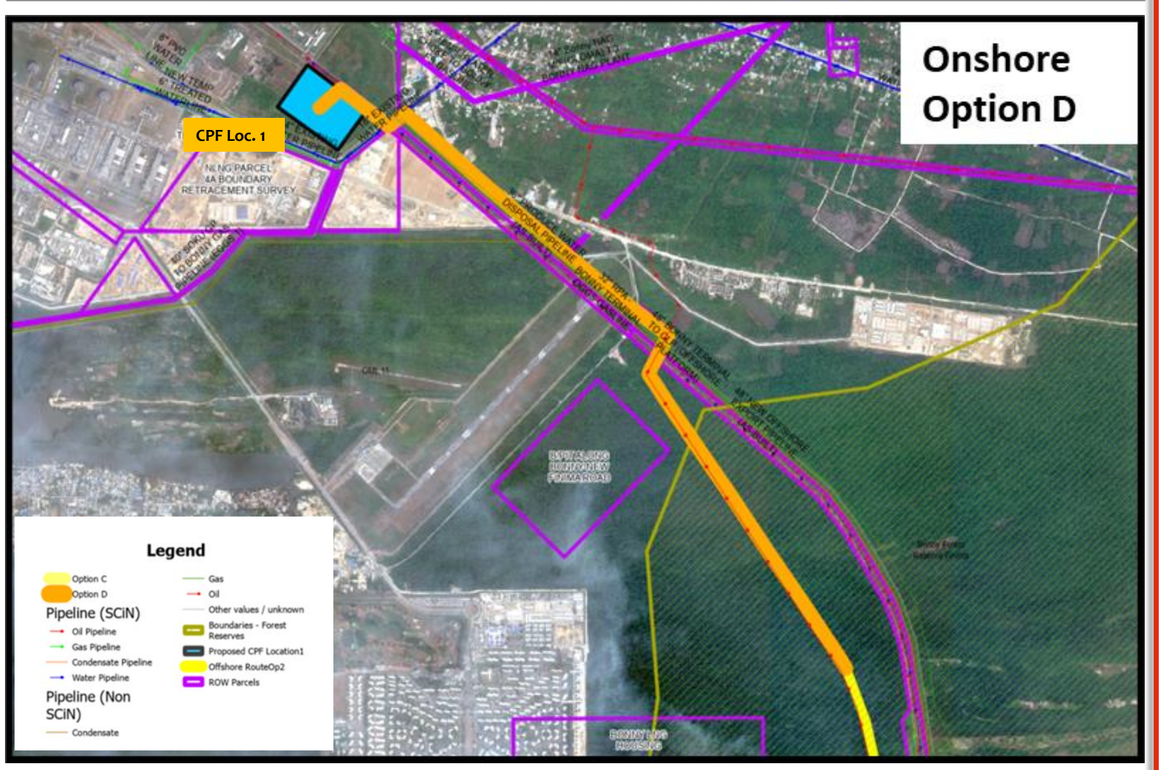
OBJECTIVE:
To identify and propose onshore gas-line route options for Gas evacuation starting from proposed offshore gas-line route option landfall from Nnwadoro field location (OML 135) to NLNG plant in BOGT.

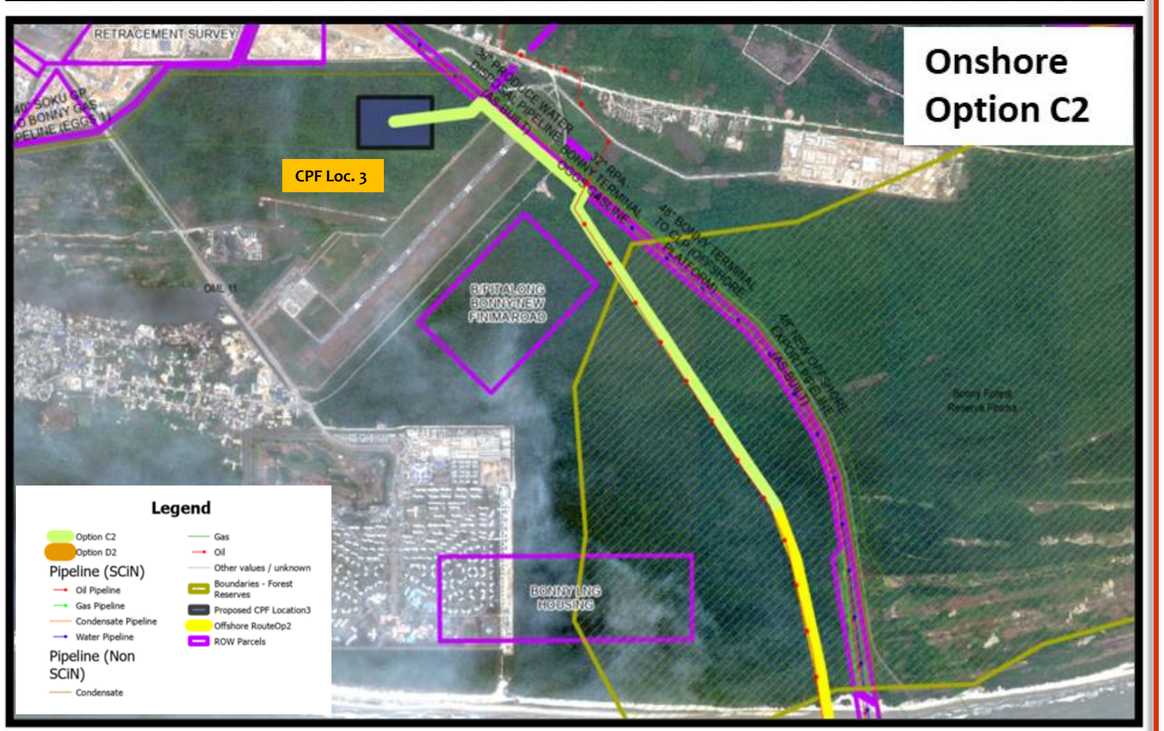
SUMMARY:
From the Route Assessment Statistics for offshore landfall route option 2, a total of four (4) proposed onshore route options were investigated to two (2) proposed CPF locations (2 options per CPF location) in BOGT field location. Proposed Route Option 2C has the shortest route length of 168.16km to CPF location 1 while Proposed Route Option 2C2 has the shortest route length of 166.79km to CPF location 3.

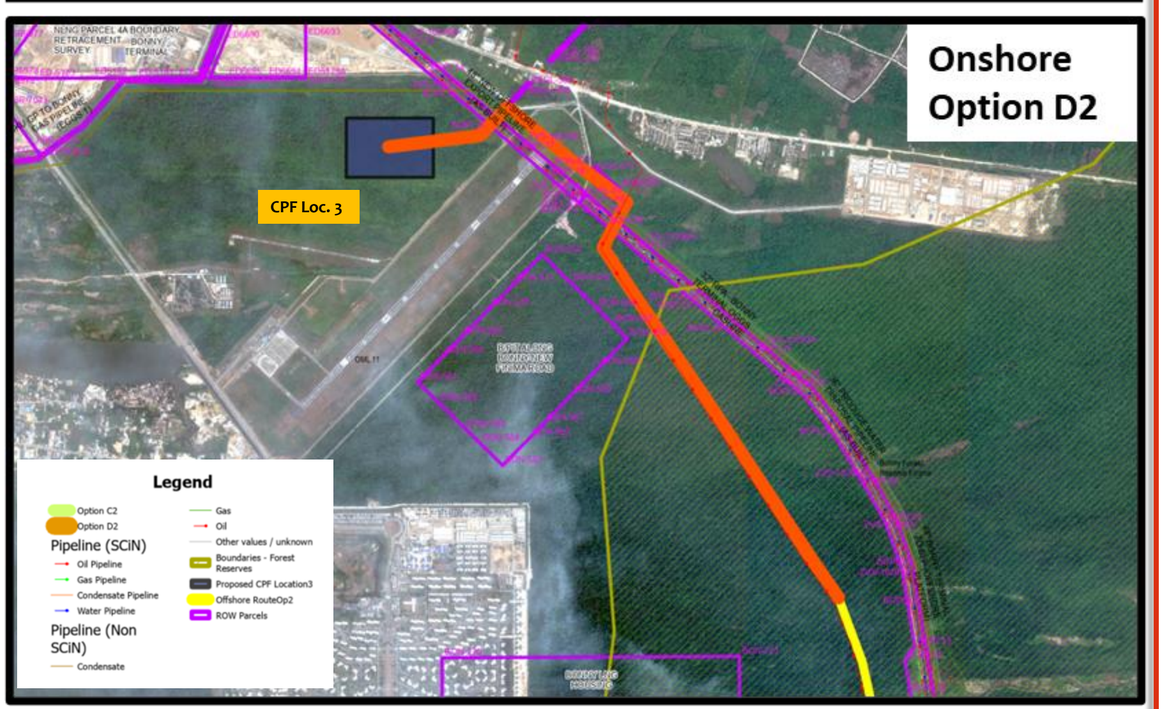
METHODOLOGY:
- Analysis of the most recent high-res satellite imagery for land-use and land cover.
- Considerations for planning the route are environmental, technical, economic and socio-cultural issues using satellite imagery and vector data.
- Generate statistical assessment for the proposed gas line route.

DATASETS USED:
Raster- High Resolution Satellite Imagery of 0.5m resolution acquired January 2023 (Source: SPDC Geomatics SDE).
Vector- The dataset used from Geomatics corporate database are: SPDC Land Parcels Portfolio; Settlement layer data; Electric Powerlines data; Pipelines(SCiN); Flowlines(SCiN); Telecoms All; Gas Plants; Flow Station and Manifold layers.









ROUTE ASSESSMENT STATISTICS FOR NNWADORO PIPELINE

DESCRIPTION	Route Option 2 to CPF Location 1		Route Option 2 to CPF Location 3	
	Option 2C	Option 2D	Option 2C2	Option 2D2
***Approx. total length (km)	168.16	168.31	166.79	167.02
Length Offshore	163.2	163.2	163.2	163.2
***Length Onshore	4.96	5.11	3.59	3.82
***Length through existing SPDC ROW (km)	0.67	0.68	0.83	0.11
***Length through new ROW (km)	4.29	4.43	2.76	3.71
***No. of Forest reserve traversed	1	1	1	1
***No of Communities within 2km buffer	5 (NAMATA, AYAMBO, OGUGUMANGA, AGAJA, BONNY)	5 (NAMATA, AYAMBO, OGUGUMANGA, AGAJA, BONNY)	4 (NAMATA, OGUGUMANGA, AGAJA, BONNY)	4 (NAMATA, OGUGUMANGA, AGAJA, BONNY)
***No. of existing river/creek crossing	Nil	Nil	Nil	Nil
***Width range of river /creek crossings (m)	Nil	Nil	Nil	Nil
***No river /creek crossing with width of less 20m	Nil	Nil	Nil	Nil
***No river /creek crossing with width Greater than 20m	Nil	Nil	Nil	Nil
***No river /creek crossing with width Greater than 50m	Nil	Nil	Nil	Nil
***No river /creek crossing with width Greater than 100m	Nil	Nil	Nil	Nil
No of Flowline in RoW Corridor	Nil	Nil	Nil	Nil
***Pipeline crossings	Several	Several	Several	Several
***No of Telecom(Nitel) cable crossings	Nil	Nil	Nil	Nil
***No of Electrical cable crossings	Nil	Nil	Nil	Nil
***Approx. Length of Farmland traversed (Km)	Nil	Nil	Nil	Nil
***Approx. Length of Impacted Vegetation traversed (Km)	4.96	5.11	3.59	3.82
Approx. Length of Mature Forest traversed (Km)	Nil	Nil	Nil	Nil
Approx. Length of Swamp Vegetation traversed (Km)	Nil	Nil	Nil	Nil
***Approx. Length of Settlement traversed (Km)	2.57	2.73	1.34	1.43
***Major Road Crossing	3	3	1	1

CONCLUSION:
From the route assessment statistics for offshore landfall route option 2, four (4) proposed onshore route options were investigated to two (2) proposed CPF (1 and 3) locations (2 options per CPF location) in BOGT field location for this project. As required, several vital statistics based on the Environmental and Safety, Engineering and Operational, Biodiversity considerations were analyzed and extracted in the Route Assessment Statistical table above. The route options were not ranked in this study but left for ranking during the multi discipline route selection workshop using the SCAN tool. The descriptions on the route assessment statistics table marked (***) are part of the criteria to be used with the SCAN tool for ranking the identified route options during the route selection workshop.

PREPARED BY: Amaechi Ojeh (SPDC-UPV/C/NGI) **CLIENT: Tarryla Nnoaham (SPDC-UPV/C/NF)** **ONSHORE OPs REP: Seember Gbargbar (SPDC-UPV/C/NGO)** **REVIEWED BY: Simisola Oyetunde (SPDC-UPV/C/NGI)**

DATE: 2nd June 2023

ROUTE INVESTIGATION REPORT FOR NNWADORO PIPELINE PROJECT (ROUTE OPTION 5)

INTRODUCTION:
This project has an opportunity to secure a custom-made financial plan with NNPC to create an investable "Nigeria first" deep water gas development with ~9Tcf of Nnwa-Doro gas and safely supply the NLNG plant with ~700MMscf/d for 20years from 2029. This aligns with the government's decade of gas aspirations and creates immense value to Nigeria and unit parties.

OBJECTIVE:
To identify and propose onshore gas-line route options for Gas evacuation starting from proposed offshore gas-line route option landfall from Nnwadoro field location (OML 135) to NLNG plant in BOGT.

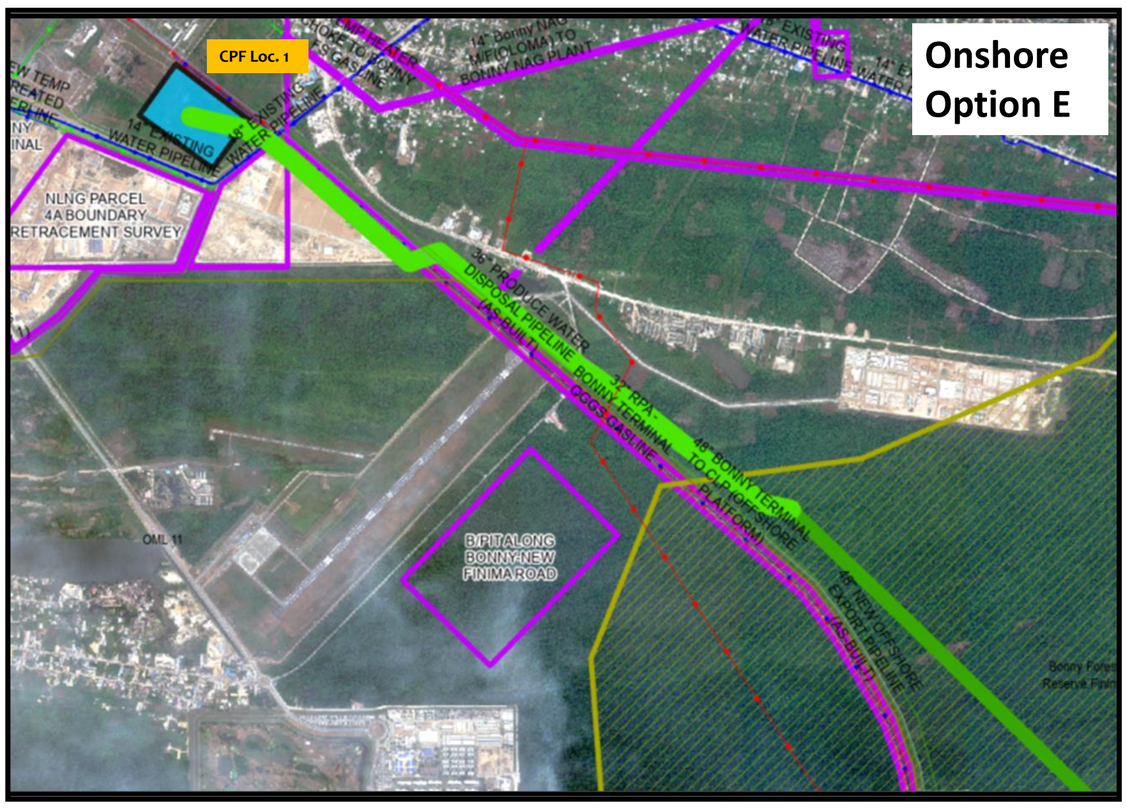
SUMMARY:
From the Route Assessment Statistics for offshore landfall route option 5, a total of two (2) proposed onshore route options were investigated to two (2) proposed CPF locations (1 option per CPF location) in BOGT field location.

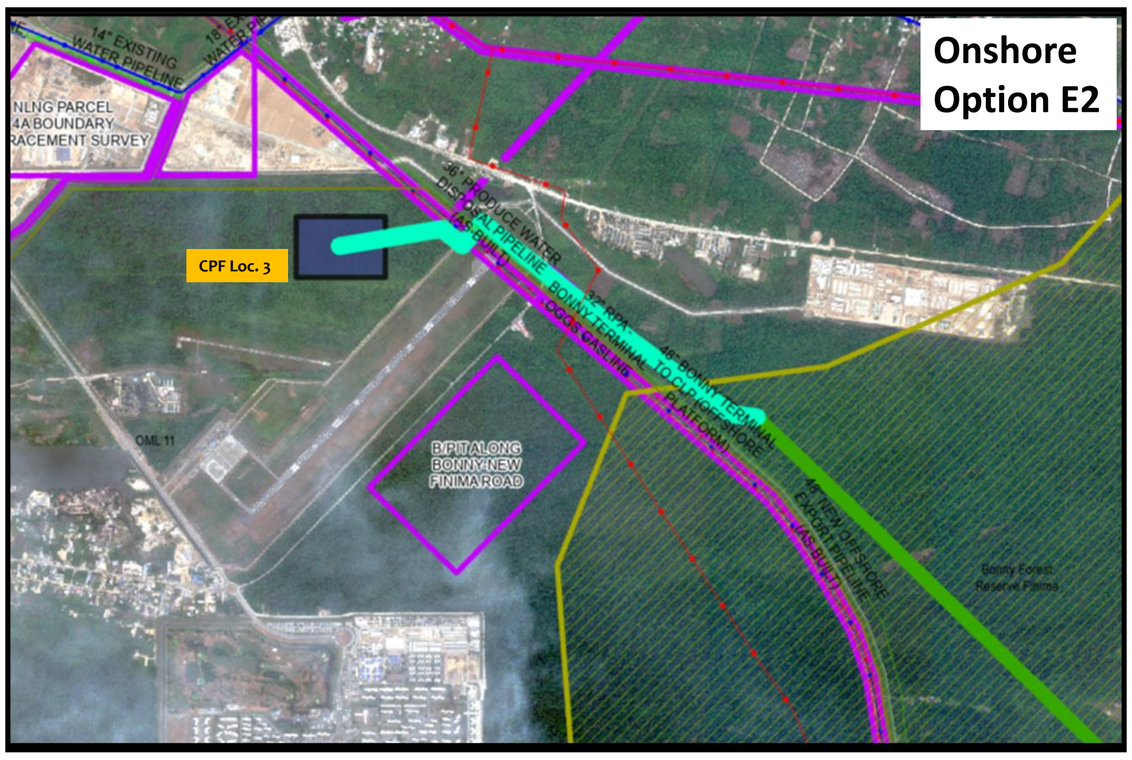
METHODOLOGY:
- Analysis of the most recent high-res satellite imagery for land-use and land cover.
- Considerations for planning the route are environmental, technical, economic and socio-cultural issues using satellite imagery and vector data.
- Generate statistical assessment for the proposed gas line route.

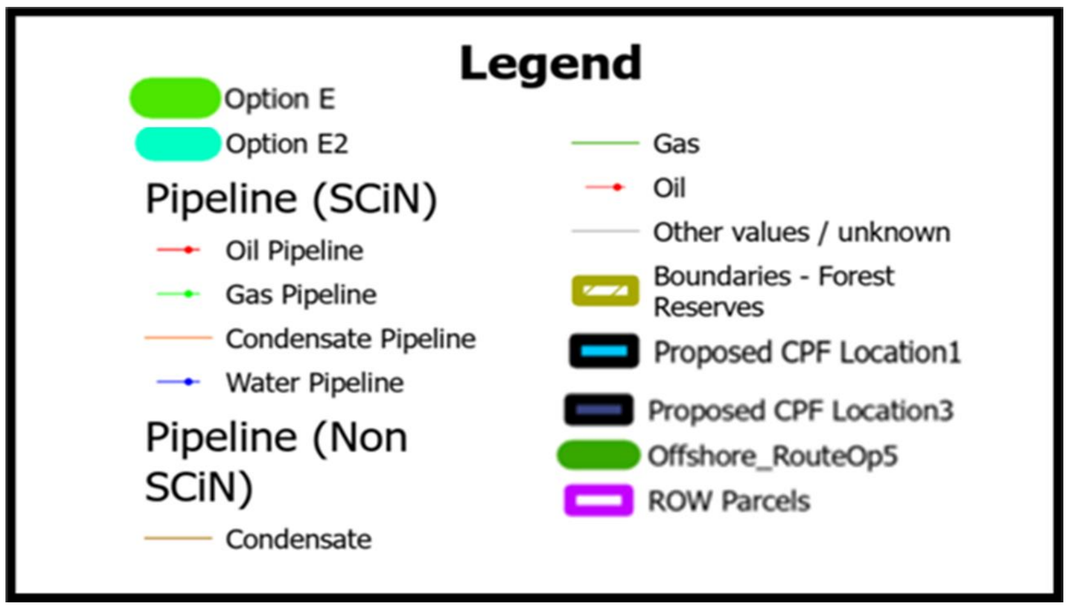
DATASETS USED:
Raster- High Resolution Satellite Imagery of 0.5m resolution acquired January 2023 (Source: SPDC Geomatics SDE).
Vector- The dataset used from Geomatics corporate database are: SPDC Land Parcels Portfolio; Settlement layer data; Electric Powerlines data; Pipelines(SCiN); Flowlines(SCiN); Telecoms All; Gas Plants; Flow Station and Manifold layers.

ROUTE ASSESSMENT STATISTICS FOR NNWADORO PIPELINE

DESCRIPTION	Route Option 5 to CPF Location 1	Route Option 5 to CPF Location 3
	Option 5E	Option 5E2
***Approx. total length (km)	190.42	189.26
Length Offshore	183.42	183.42
***Length Onshore	7.00	5.84
***Length through existing SPDC ROW (km)	0.65	0.06
***Length through new ROW (km)	6.35	5.78
***No. of Forest reserve traversed	1	1
***No of Communities within 2km buffer	3 (AYAMBO, OGUGUMANGA, BONNY)	2 (OGUGUMANGA, BONNY)
***No. of existing river/creek crossing	Nil	Nil
***Width range of river /creek crossings (m)	Nil	Nil
***No river /creek crossing with width of less 20m	Nil	Nil
***No river /creek crossing with width Greater than 20m	Nil	Nil
***No river /creek crossing with width Greater than 50m	Nil	Nil
***No river /creek crossing with width Greater than 100m	Nil	Nil
No of Flowline in RoW Corridor	Nil	Nil
***Pipeline crossings	Several	Several
***No of Telecom(Nitel) cable crossings	Nil	Nil
***No of Electrical cable crossings	Nil	Nil
***Approx. Length of Farmland traversed (Km)	Nil	Nil
***Approx. Length of Impacted Vegetation traversed (Km)	7.00	5.84
Approx. Length of Mature Forest traversed (Km)	Nil	Nil
Approx. Length of Swamp Vegetation traversed (Km)	Nil	Nil
***Approx. Length of Settlement traversed (Km)	3.62	2.46
***Major Road Crossing	3	1







CONCLUSION:
From the route assessment statistics for offshore landfall route option 5, two (2) proposed onshore route options were investigated to two (2) proposed CPF (1 and 3) locations (1 option per CPF location) in BOGT field location for this project. As required, several vital statistics based on the Environmental and Safety, Engineering and Operational, Biodiversity considerations were analyzed and extracted in the Route Assessment Statistical table above. The route options were not ranked in this study but left for ranking during the multi discipline route selection workshop using the SCAN tool. The descriptions on the route assessment statistics table marked (***) are part of the criteria to be used with the SCAN tool for ranking the identified route options during the route selection workshop.

PREPARED BY: Amaechi Ojeh (SPDC-UPV/C/NGI) **CLIENT: Tarryla Nnoaham (SPDC-UPV/C/NF)** **ONSHORE OPs REP: Seember Gbargbar (SPDC-UPV/C/NGO)** **REVIEWED BY: Simisola Oyetunde (SPDC-UPV/C/NGI)**

DATE: 2nd June 2023

ROUTE INVESTIGATION REPORT FOR NNWADORO PIPELINE PROJECT (ROUTE OPTION 4)

INTRODUCTION:
This project has an opportunity to secure a custom-made financial plan with NNPC to create an investable "Nigeria first" deep water gas development with ~9Tcf of Nnwa-Doro gas and safely supply the NLNG plant with ~700MMscf/d for 20years from 2029. This aligns with the government's decade of gas aspirations and creates immense value to Nigeria and unit parties.

OBJECTIVE:
To identify and propose onshore gas-line route options for Gas evacuation starting from proposed offshore gas-line route option landfall from Nnwadoro field location (OML 135) to NLNG plant in BOGT.

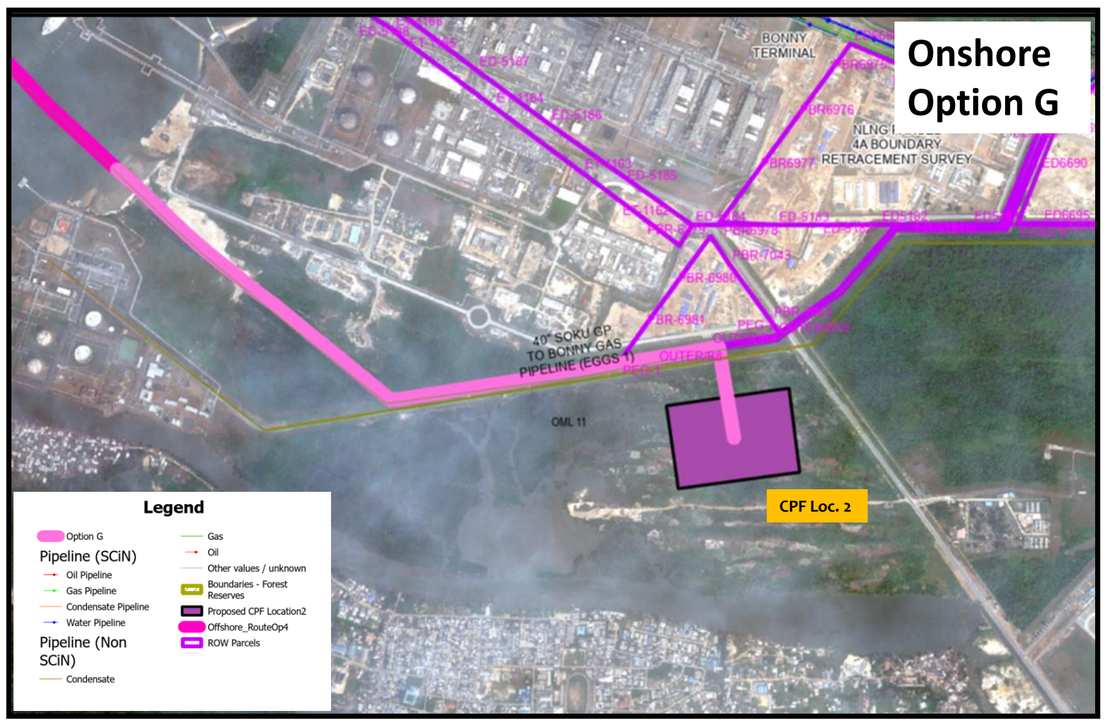
SUMMARY:
From the Route Assessment Statistics for offshore landfall route option 4, a total of three (3) proposed onshore route options were investigated to three (3) proposed CPF locations (1 option per CPF location) in BOGT field location.

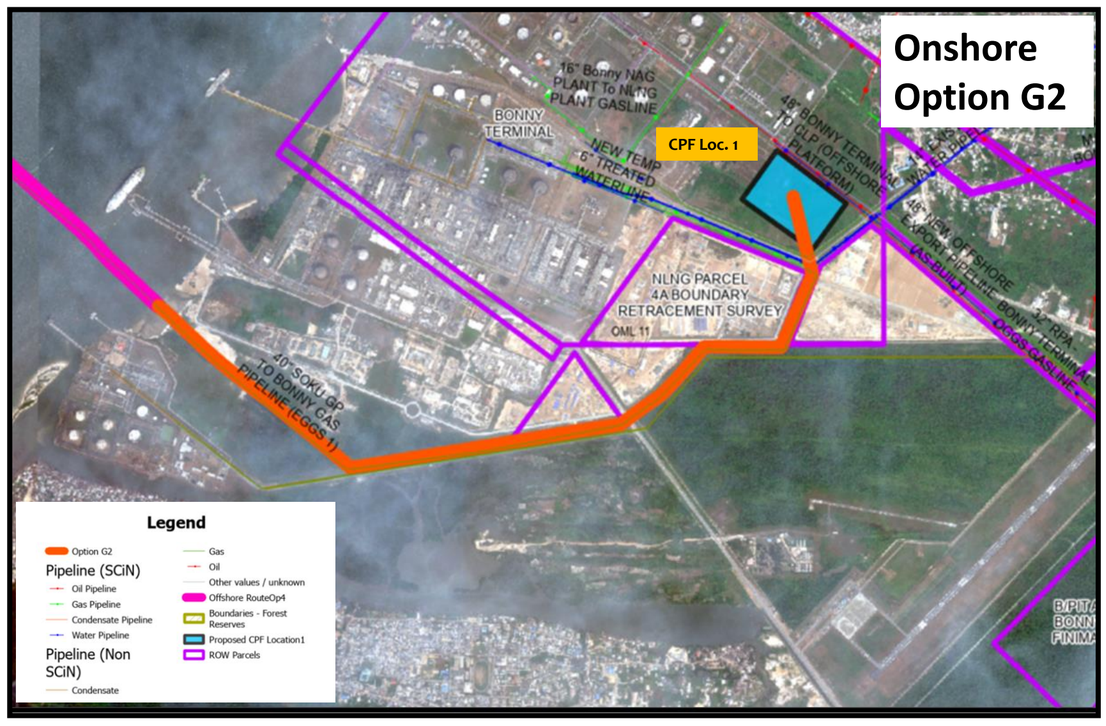
METHODOLOGY:
- Analysis of the most recent high-res satellite imagery for land-use and land cover.
- Considerations for planning the route are environmental, technical, economic and socio-cultural issues using satellite imagery and vector data.
- Generate statistical assessment for the proposed gas line route.

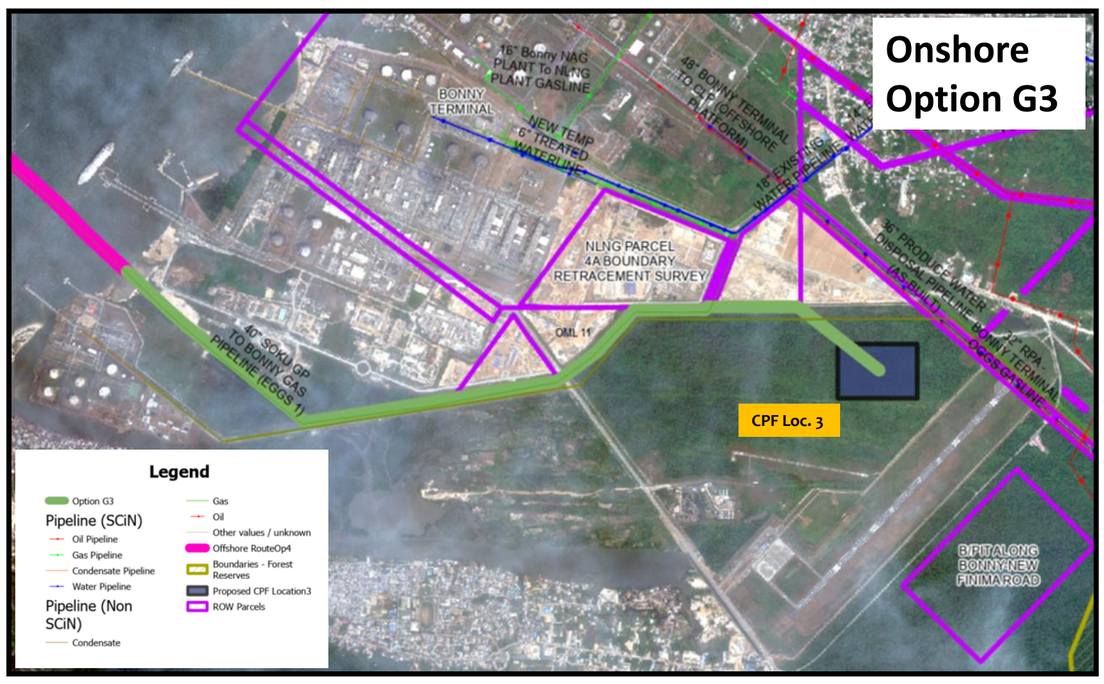
DATASETS USED:
Raster- High Resolution Satellite Imagery of 0.5m resolution acquired January 2023 (Source: SPDC Geomatics SDE).
Vector- The dataset used from Geomatics corporate database are: SPDC Land Parcels Portfolio; Settlement layer data; Electric Powerlines data; Pipelines(SCiN); Flowlines(SCiN); Telecoms All; Gas Plants; Flow Station and Manifold layers.

ROUTE ASSESSMENT STATISTICS FOR NNWADORO PIPELINE

DESCRIPTION	Route Option 4 to CPF Location 2	Route Option 4 to CPF Location 1	Route Option 4 to CPF Location 3
	Option 4G	Option 4G2	Option 4G3
***Approx. total length (km)	166.15	167.83	168.10
Length Offshore	163.34	163.34	163.34
***Length Onshore	2.81	4.49	4.76
***Length through existing SPDC ROW (km)	2.48	4.49	4.19
***Length through new ROW (km)	0.33	Nil	0.57
***No. of Forest reserve traversed	Nil	Nil	Nil
***No of Communities within 2km buffer	2 (BONNY, FINIMA)	2 (BONNY, FINIMA)	2 (BONNY, FINIMA)
***No. of existing river/creek crossing	Nil	Nil	Nil
***Width range of river /creek crossings (m)	Nil	Nil	Nil
***No river /creek crossing with width of less 20m	Nil	Nil	Nil
***No river /creek crossing with width Greater than 20m	Nil	Nil	Nil
***No river /creek crossing with width Greater than 50m	Nil	Nil	Nil
***No river /creek crossing with width Greater than 100m	Nil	Nil	Nil
No of Flowline in RoW Corridor	Nil	Nil	Nil
***Pipeline crossings	Several	Several	Several
***No of Telecom(Nitel) cable crossings	Nil	Nil	Nil
***No of Electrical cable crossings	Nil	Nil	Nil
***Approx. Length of Farmland traversed (Km)	Nil	Nil	Nil
***Approx. Length of Impacted Vegetation traversed (Km)	2.81	4.49	4.76
Approx. Length of Mature Forest traversed (Km)	Nil	Nil	Nil
Approx. Length of Swamp Vegetation traversed (Km)	Nil	Nil	Nil
***Approx. Length of Settlement traversed (Km)	2.81	4.49	4.76
***Major Road Crossing	Nil	4	2







CONCLUSION:
From the route assessment statistics for offshore landfall route option 4, three (3) proposed onshore route options were investigated to three (3) proposed CPF (1, 2 and 3) locations (1 option per CPF location) in BOGT field location for this project. As required, several vital statistics based on the Environmental and Safety, Engineering and Operational, Biodiversity considerations were analyzed and extracted in the Route Assessment Statistical table above. The route options were not ranked in this study but left for ranking during the multi discipline route selection workshop using the SCAN tool. The descriptions on the route assessment statistics table marked (***) are part of the criteria to be used with the SCAN tool for ranking the identified route options during the route selection workshop.

PREPARED BY: Amaechi Ojeh (SPDC-UPV/C/NGI) **CLIENT: Tarryla Nnoaham (SPDC-UPV/C/NF)** **ONSHORE OPs REP: Seember Gbargbar (SPDC-UPV/C/NGO)** **REVIEWED BY: Simisola Oyetunde (SPDC-UPV/C/NGI)**

DATE: 2nd June 2023